

SUPPLEMENTARY MATERIAL: THE CERTIFICATE TABLES FOR “A DICHOTOMY FOR ERDŐS PROBLEM 1112”

JOHAN LAND

This supplement to the paper *A dichotomy for Erdős problem 1112: existence of $r_k(d_1, d_2)$ for $k \geq 3$* lists in full the finite certificate tables referenced in its appendix on the certificate tables. Each entry is a hard-core triple (a, b, M) together with a box certificate (x, Y, Z) , asserting that the multiset of x copies of a , Y of b and Z of M has budget $x + Y + Z \leq M - 1$ and subset sums containing M consecutive integers; see the main paper for the definitions, the two-generator/frame mechanism, and the reconciliation of printed rows against the rows decided by the Lean kernel.

These tables are generated mechanically from the canonical data (`data/certificate-data.md`) by `gen-tables.py`, which re-verifies every row — budget $\leq M - 1$ and coverage of M consecutive integers, by exact arbitrary-precision bitmask — at generation time, aborting on any failure. The same rows are transcribed into the Lean 4 development and decided by the kernel, and are exported in machine-readable form as `data/table-A.csv` and `data/table-B.csv` with SHA-256 digests in `data/SHA256SUMS`. Source, proof, and data: github.com/beetree/math_erdos_1112.

Table A: The 172 exceptional Case-T triples (the T-tail lemma of the main paper), all with $a \leq 29$; the scan over $a \leq 3000$ finds no others. Format $(a, b, M) \rightarrow (x, Y, Z)$: the multiset of x copies of a , Y of b , Z of M .

$(4, 9, 10) \rightarrow (7, 1, 1)$	$(4, 13, 14) \rightarrow (10, 1, 1)$	$(5, 8, 9) \rightarrow (5, 2, 1)$	$(5, 9, 12) \rightarrow (7, 2, 1)$
$(5, 11, 12) \rightarrow (7, 1, 2)$	$(5, 11, 13) \rightarrow (8, 2, 1)$	$(5, 12, 14) \rightarrow (9, 1, 2)$	$(5, 13, 14) \rightarrow (8, 2, 1)$
$(5, 13, 16) \rightarrow (10, 1, 2)$	$(6, 11, 14) \rightarrow (9, 1, 2)$	$(6, 11, 15) \rightarrow (9, 2, 1)$	$(6, 13, 14) \rightarrow (9, 1, 2)$
$(6, 13, 15) \rightarrow (10, 2, 1)$	$(6, 13, 16) \rightarrow (10, 1, 2)$	$(7, 10, 12) \rightarrow (7, 3, 1)$	$(7, 10, 15) \rightarrow (8, 2, 2)$
$(7, 11, 12) \rightarrow (7, 2, 2)$	$(7, 11, 13) \rightarrow (7, 2, 2)$	$(7, 11, 16) \rightarrow (8, 3, 1)$	$(7, 12, 13) \rightarrow (7, 3, 1)$
$(7, 12, 15) \rightarrow (8, 2, 2)$	$(7, 13, 17) \rightarrow (9, 3, 1)$	$(7, 13, 18) \rightarrow (9, 2, 2)$	$(7, 15, 16) \rightarrow (9, 1, 3)$
$(7, 15, 17) \rightarrow (9, 2, 2)$	$(7, 15, 18) \rightarrow (10, 3, 1)$	$(7, 16, 17) \rightarrow (10, 2, 2)$	$(7, 16, 18) \rightarrow (11, 1, 3)$
$(8, 11, 12) \rightarrow (7, 3, 1)$	$(8, 11, 15) \rightarrow (9, 2, 2)$	$(8, 13, 14) \rightarrow (7, 3, 2)$	$(8, 13, 15) \rightarrow (8, 2, 2)$
$(8, 13, 17) \rightarrow (10, 2, 2)$	$(8, 15, 18) \rightarrow (9, 3, 2)$	$(8, 15, 19) \rightarrow (11, 2, 2)$	$(8, 17, 18) \rightarrow (11, 1, 3)$
$(8, 17, 19) \rightarrow (10, 2, 2)$	$(9, 13, 16) \rightarrow (8, 3, 2)$	$(9, 13, 20) \rightarrow (9, 4, 1)$	$(9, 14, 15) \rightarrow (8, 2, 2)$
$(9, 14, 16) \rightarrow (9, 4, 1)$	$(9, 14, 17) \rightarrow (9, 2, 3)$	$(9, 14, 20) \rightarrow (9, 3, 2)$	$(9, 16, 17) \rightarrow (9, 4, 1)$
$(9, 16, 19) \rightarrow (10, 3, 2)$	$(9, 17, 20) \rightarrow (11, 2, 3)$	$(9, 19, 20) \rightarrow (11, 1, 4)$	$(10, 13, 15) \rightarrow (9, 4, 1)$
$(10, 13, 18) \rightarrow (10, 3, 2)$	$(10, 13, 21) \rightarrow (9, 3, 2)$	$(10, 17, 18) \rightarrow (9, 3, 2)$	$(10, 17, 19) \rightarrow (9, 3, 2)$
$(10, 17, 21) \rightarrow (10, 2, 3)$	$(11, 14, 15) \rightarrow (7, 3, 4)$	$(11, 14, 18) \rightarrow (9, 5, 1)$	$(11, 14, 21) \rightarrow (10, 3, 2)$
$(11, 15, 16) \rightarrow (7, 3, 2)$	$(11, 15, 20) \rightarrow (9, 4, 2)$	$(11, 16, 18) \rightarrow (8, 4, 2)$	$(11, 16, 19) \rightarrow (10, 5, 1)$
$(11, 16, 20) \rightarrow (9, 3, 2)$	$(11, 17, 18) \rightarrow (8, 2, 3)$	$(11, 18, 19) \rightarrow (9, 4, 3)$	$(11, 18, 20) \rightarrow (11, 5, 1)$
$(11, 18, 21) \rightarrow (10, 2, 3)$	$(11, 19, 20) \rightarrow (9, 4, 2)$	$(11, 20, 21) \rightarrow (11, 5, 1)$	$(12, 17, 18) \rightarrow (10, 5, 1)$
$(12, 17, 20) \rightarrow (10, 3, 2)$	$(12, 17, 21) \rightarrow (10, 2, 3)$	$(12, 17, 23) \rightarrow (12, 4, 2)$	$(12, 19, 20) \rightarrow (10, 3, 2)$
$(12, 19, 21) \rightarrow (11, 2, 3)$	$(12, 19, 22) \rightarrow (11, 3, 4)$	$(12, 19, 23) \rightarrow (11, 4, 2)$	$(13, 16, 21) \rightarrow (10, 6, 1)$
$(13, 16, 24) \rightarrow (11, 2, 4)$	$(13, 17, 20) \rightarrow (9, 4, 2)$	$(13, 17, 24) \rightarrow (10, 4, 4)$	$(13, 18, 19) \rightarrow (9, 4, 2)$
$(13, 18, 20) \rightarrow (9, 3, 3)$	$(13, 18, 22) \rightarrow (11, 6, 1)$	$(13, 18, 24) \rightarrow (11, 3, 3)$	$(13, 19, 24) \rightarrow (10, 3, 3)$
$(13, 20, 21) \rightarrow (10, 2, 4)$	$(13, 20, 22) \rightarrow (10, 4, 2)$	$(13, 20, 23) \rightarrow (12, 6, 1)$	$(13, 20, 24) \rightarrow (11, 4, 2)$
$(13, 21, 22) \rightarrow (10, 4, 4)$	$(13, 21, 24) \rightarrow (11, 2, 4)$	$(13, 22, 23) \rightarrow (11, 3, 3)$	$(13, 22, 24) \rightarrow (13, 6, 1)$
$(13, 23, 24) \rightarrow (11, 4, 2)$	$(14, 17, 24) \rightarrow (12, 5, 2)$	$(14, 19, 20) \rightarrow (9, 3, 3)$	$(14, 19, 21) \rightarrow (12, 6, 1)$
$(14, 19, 22) \rightarrow (9, 5, 2)$	$(14, 19, 25) \rightarrow (10, 4, 2)$	$(14, 23, 24) \rightarrow (12, 3, 5)$	$(14, 23, 25) \rightarrow (11, 3, 4)$

Date: July 5, 2026.

Table A: (continued).

$(15, 19, 24) \rightarrow (11, 2, 4)$	$(15, 22, 24) \rightarrow (11, 2, 4)$	$(15, 22, 25) \rightarrow (11, 4, 2)$	$(15, 22, 26) \rightarrow (13, 7, 1)$
$(15, 23, 24) \rightarrow (11, 2, 4)$	$(15, 23, 26) \rightarrow (12, 5, 4)$	$(16, 19, 27) \rightarrow (13, 6, 2)$	$(16, 21, 23) \rightarrow (10, 5, 2)$
$(16, 21, 24) \rightarrow (13, 7, 1)$	$(16, 21, 25) \rightarrow (10, 4, 3)$	$(16, 23, 24) \rightarrow (13, 7, 1)$	$(16, 23, 26) \rightarrow (11, 5, 2)$
$(16, 23, 27) \rightarrow (12, 3, 4)$	$(16, 25, 26) \rightarrow (10, 3, 4)$	$(16, 25, 27) \rightarrow (12, 2, 5)$	$(17, 21, 28) \rightarrow (11, 3, 4)$
$(17, 22, 23) \rightarrow (10, 5, 6)$	$(17, 22, 25) \rightarrow (10, 4, 3)$	$(17, 22, 26) \rightarrow (10, 6, 2)$	$(17, 22, 28) \rightarrow (13, 8, 1)$
$(17, 23, 24) \rightarrow (10, 3, 4)$	$(17, 24, 25) \rightarrow (10, 5, 2)$	$(17, 24, 26) \rightarrow (11, 4, 4)$	$(17, 24, 28) \rightarrow (11, 3, 4)$
$(17, 25, 28) \rightarrow (11, 6, 2)$	$(17, 26, 27) \rightarrow (11, 2, 5)$	$(17, 26, 28) \rightarrow (11, 4, 3)$	$(17, 27, 28) \rightarrow (12, 4, 3)$
$(18, 23, 27) \rightarrow (14, 8, 1)$	$(18, 23, 29) \rightarrow (12, 4, 3)$	$(18, 25, 26) \rightarrow (10, 5, 3)$	$(18, 25, 27) \rightarrow (15, 8, 1)$
$(18, 25, 28) \rightarrow (11, 3, 4)$	$(19, 24, 30) \rightarrow (11, 5, 3)$	$(19, 25, 30) \rightarrow (11, 4, 3)$	$(19, 26, 27) \rightarrow (10, 4, 4)$
$(19, 26, 28) \rightarrow (11, 3, 4)$	$(19, 26, 29) \rightarrow (12, 3, 4)$	$(19, 26, 30) \rightarrow (12, 6, 2)$	$(19, 27, 28) \rightarrow (12, 6, 2)$
$(19, 28, 30) \rightarrow (13, 3, 5)$	$(19, 29, 30) \rightarrow (13, 2, 6)$	$(20, 27, 28) \rightarrow (11, 3, 4)$	$(20, 27, 29) \rightarrow (12, 6, 2)$
$(20, 27, 30) \rightarrow (16, 9, 1)$	$(20, 27, 31) \rightarrow (12, 5, 4)$	$(20, 29, 30) \rightarrow (16, 9, 1)$	$(21, 29, 30) \rightarrow (10, 5, 4)$
$(21, 29, 32) \rightarrow (12, 3, 5)$	$(22, 29, 31) \rightarrow (11, 6, 3)$	$(22, 29, 32) \rightarrow (11, 5, 4)$	$(22, 29, 33) \rightarrow (17, 10, 1)$
$(22, 31, 32) \rightarrow (12, 5, 3)$	$(22, 31, 33) \rightarrow (18, 10, 1)$	$(23, 30, 31) \rightarrow (13, 7, 8)$	$(23, 30, 33) \rightarrow (13, 7, 2)$
$(23, 30, 34) \rightarrow (13, 4, 4)$	$(23, 31, 32) \rightarrow (11, 3, 5)$	$(23, 31, 34) \rightarrow (11, 6, 4)$	$(23, 32, 33) \rightarrow (13, 3, 5)$
$(23, 32, 34) \rightarrow (12, 4, 4)$	$(23, 33, 34) \rightarrow (13, 7, 2)$	$(24, 31, 35) \rightarrow (12, 4, 4)$	$(25, 33, 36) \rightarrow (14, 8, 2)$
$(25, 34, 35) \rightarrow (13, 4, 4)$	$(25, 34, 36) \rightarrow (13, 3, 6)$	$(26, 35, 36) \rightarrow (13, 3, 6)$	$(26, 35, 37) \rightarrow (13, 6, 3)$
$(27, 37, 38) \rightarrow (14, 6, 3)$	$(28, 37, 39) \rightarrow (14, 4, 5)$	$(29, 38, 39) \rightarrow (15, 9, 10)$	$(29, 39, 40) \rightarrow (14, 3, 7)$

Table B: The 178 Case-B class bases ($a \leq 11$, $\mu \geq 12$). Format $[a; \bar{e}, h]$ followed by the base triple $(a, b, M) \rightarrow (x, Y, Z)$. The λ -lift lemma of the main paper extends each certificate to every member of its class with larger e .

$[4; 1, 1] (4, 17, 18) \rightarrow (13, 1, 1)$	$[5; 1, 1] (5, 16, 17) \rightarrow (10, 1, 2)$	$[5; 1, 2] (5, 16, 18) \rightarrow (11, 2, 1)$
$[5; 2, 2] (5, 17, 19) \rightarrow (12, 1, 2)$	$[5; 3, 1] (5, 18, 19) \rightarrow (11, 2, 1)$	$[5; 3, 3] (5, 18, 21) \rightarrow (13, 1, 2)$
$[5; 4, 3] (5, 14, 17) \rightarrow (10, 2, 1)$	$[6; 1, 1] (6, 19, 20) \rightarrow (13, 1, 2)$	$[6; 1, 2] (6, 19, 21) \rightarrow (14, 2, 1)$
$[6; 1, 3] (6, 19, 22) \rightarrow (14, 1, 2)$	$[6; 5, 3] (6, 17, 20) \rightarrow (13, 1, 2)$	$[6; 5, 4] (6, 17, 21) \rightarrow (13, 2, 1)$
$[7; 1, 1] (7, 22, 23) \rightarrow (13, 1, 3)$	$[7; 1, 2] (7, 22, 24) \rightarrow (13, 2, 2)$	$[7; 1, 3] (7, 22, 25) \rightarrow (14, 3, 1)$
$[7; 1, 4] (7, 15, 19) \rightarrow (11, 2, 2)$	$[7; 2, 1] (7, 23, 24) \rightarrow (14, 2, 2)$	$[7; 2, 2] (7, 23, 25) \rightarrow (15, 1, 3)$
$[7; 2, 4] (7, 16, 20) \rightarrow (11, 2, 2)$	$[7; 3, 2] (7, 17, 19) \rightarrow (11, 3, 1)$	$[7; 3, 3] (7, 17, 20) \rightarrow (12, 1, 3)$
$[7; 3, 5] (7, 17, 22) \rightarrow (12, 2, 2)$	$[7; 4, 1] (7, 18, 19) \rightarrow (11, 2, 2)$	$[7; 4, 2] (7, 18, 20) \rightarrow (11, 2, 2)$
$[7; 4, 4] (7, 18, 22) \rightarrow (13, 1, 3)$	$[7; 4, 5] (7, 18, 23) \rightarrow (12, 3, 1)$	$[7; 5, 1] (7, 19, 20) \rightarrow (11, 3, 1)$
$[7; 5, 3] (7, 19, 22) \rightarrow (12, 2, 2)$	$[7; 5, 5] (7, 19, 24) \rightarrow (14, 1, 3)$	$[7; 6, 3] (7, 20, 23) \rightarrow (13, 2, 2)$
$[7; 6, 4] (7, 20, 24) \rightarrow (13, 3, 1)$	$[7; 6, 5] (7, 20, 25) \rightarrow (13, 2, 2)$	$[8; 1, 1] (8, 25, 26) \rightarrow (16, 1, 3)$
$[8; 1, 2] (8, 25, 27) \rightarrow (14, 2, 2)$	$[8; 1, 3] (8, 17, 20) \rightarrow (12, 3, 1)$	$[8; 1, 4] (8, 17, 21) \rightarrow (12, 2, 2)$
$[8; 1, 5] (8, 17, 22) \rightarrow (11, 3, 2)$	$[8; 3, 1] (8, 19, 20) \rightarrow (12, 3, 1)$	$[8; 3, 3] (8, 19, 22) \rightarrow (14, 1, 3)$
$[8; 3, 4] (8, 19, 23) \rightarrow (14, 2, 2)$	$[8; 3, 6] (8, 19, 25) \rightarrow (12, 2, 2)$	$[8; 5, 1] (8, 21, 22) \rightarrow (11, 3, 2)$
$[8; 5, 2] (8, 21, 23) \rightarrow (12, 2, 2)$	$[8; 5, 4] (8, 21, 25) \rightarrow (15, 2, 2)$	$[8; 5, 5] (8, 21, 26) \rightarrow (16, 1, 3)$
$[8; 7, 3] (8, 23, 26) \rightarrow (13, 3, 2)$	$[8; 7, 4] (8, 23, 27) \rightarrow (16, 2, 2)$	$[8; 7, 5] (8, 15, 20) \rightarrow (11, 3, 1)$
$[8; 7, 6] (8, 15, 21) \rightarrow (10, 2, 2)$	$[9; 1, 1] (9, 28, 29) \rightarrow (16, 1, 4)$	$[9; 1, 2] (9, 19, 21) \rightarrow (12, 2, 2)$
$[9; 1, 3] (9, 19, 22) \rightarrow (12, 3, 2)$	$[9; 1, 4] (9, 19, 23) \rightarrow (12, 4, 1)$	$[9; 1, 5] (9, 19, 24) \rightarrow (13, 2, 2)$
$[9; 1, 6] (9, 19, 25) \rightarrow (12, 4, 2)$	$[9; 2, 1] (9, 20, 21) \rightarrow (12, 2, 2)$	$[9; 2, 2] (9, 20, 22) \rightarrow (13, 1, 4)$
$[9; 2, 3] (9, 20, 23) \rightarrow (13, 2, 3)$	$[9; 2, 4] (9, 20, 24) \rightarrow (13, 2, 2)$	$[9; 2, 6] (9, 20, 26) \rightarrow (13, 3, 2)$
$[9; 4, 2] (9, 22, 24) \rightarrow (13, 2, 2)$	$[9; 4, 3] (9, 22, 25) \rightarrow (13, 3, 2)$	$[9; 4, 4] (9, 22, 26) \rightarrow (15, 1, 4)$
$[9; 4, 6] (9, 22, 28) \rightarrow (13, 4, 2)$	$[9; 4, 7] (9, 22, 29) \rightarrow (14, 4, 1)$	$[9; 5, 1] (9, 23, 24) \rightarrow (13, 2, 2)$
$[9; 5, 2] (9, 23, 25) \rightarrow (14, 4, 1)$	$[9; 5, 3] (9, 23, 26) \rightarrow (14, 2, 3)$	$[9; 5, 5] (9, 23, 28) \rightarrow (16, 1, 4)$
$[9; 5, 6] (9, 23, 29) \rightarrow (14, 3, 2)$	$[9; 5, 7] (9, 14, 21) \rightarrow (10, 2, 2)$	$[9; 7, 1] (9, 25, 26) \rightarrow (14, 4, 1)$
$[9; 7, 3] (9, 25, 28) \rightarrow (15, 3, 2)$	$[9; 7, 5] (9, 16, 21) \rightarrow (11, 2, 2)$	$[9; 7, 6] (9, 16, 22) \rightarrow (10, 4, 2)$
$[9; 7, 7] (9, 25, 32) \rightarrow (18, 1, 4)$	$[9; 8, 3] (9, 26, 29) \rightarrow (16, 2, 3)$	$[9; 8, 4] (9, 17, 21) \rightarrow (11, 2, 2)$
$[9; 8, 5] (9, 17, 22) \rightarrow (11, 4, 1)$	$[9; 8, 6] (9, 17, 23) \rightarrow (11, 3, 2)$	$[9; 8, 7] (9, 17, 24) \rightarrow (12, 2, 2)$
$[10; 1, 1] (10, 21, 22) \rightarrow (13, 1, 4)$	$[10; 1, 2] (10, 21, 23) \rightarrow (11, 2, 3)$	$[10; 1, 3] (10, 21, 24) \rightarrow (11, 3, 2)$

Table B: *(continued)*.

$[10; 1, 4] (10, 21, 25) \rightarrow (14, 4, 1)$	$[10; 1, 5] (10, 21, 26) \rightarrow (14, 3, 2)$	$[10; 1, 6] (10, 21, 27) \rightarrow (13, 3, 2)$
$[10; 1, 7] (10, 21, 28) \rightarrow (14, 3, 3)$	$[10; 3, 1] (10, 23, 24) \rightarrow (12, 3, 3)$	$[10; 3, 2] (10, 23, 25) \rightarrow (15, 4, 1)$
$[10; 3, 3] (10, 23, 26) \rightarrow (16, 1, 4)$	$[10; 3, 5] (10, 23, 28) \rightarrow (16, 3, 2)$	$[10; 3, 6] (10, 23, 29) \rightarrow (14, 2, 3)$
$[10; 3, 8] (10, 23, 31) \rightarrow (14, 3, 2)$	$[10; 7, 1] (10, 27, 28) \rightarrow (14, 3, 2)$	$[10; 7, 2] (10, 27, 29) \rightarrow (14, 3, 2)$
$[10; 7, 4] (10, 27, 31) \rightarrow (15, 2, 3)$	$[10; 7, 5] (10, 17, 22) \rightarrow (12, 3, 2)$	$[10; 7, 7] (10, 27, 34) \rightarrow (20, 1, 4)$
$[10; 7, 8] (10, 17, 25) \rightarrow (12, 4, 1)$	$[10; 9, 3] (10, 19, 22) \rightarrow (11, 3, 3)$	$[10; 9, 4] (10, 19, 23) \rightarrow (11, 3, 2)$
$[10; 9, 5] (10, 19, 24) \rightarrow (13, 3, 2)$	$[10; 9, 6] (10, 19, 25) \rightarrow (13, 4, 1)$	$[10; 9, 7] (10, 19, 26) \rightarrow (11, 3, 2)$
$[10; 9, 8] (10, 19, 27) \rightarrow (12, 2, 3)$	$[11; 1, 1] (11, 23, 24) \rightarrow (13, 1, 5)$	$[11; 1, 2] (11, 23, 25) \rightarrow (12, 2, 3)$
$[11; 1, 3] (11, 23, 26) \rightarrow (12, 3, 2)$	$[11; 1, 4] (11, 23, 27) \rightarrow (13, 3, 4)$	$[11; 1, 5] (11, 23, 28) \rightarrow (14, 5, 1)$
$[11; 1, 6] (11, 23, 29) \rightarrow (14, 2, 4)$	$[11; 1, 7] (11, 23, 30) \rightarrow (13, 4, 2)$	$[11; 1, 8] (11, 23, 31) \rightarrow (14, 4, 3)$
$[11; 2, 1] (11, 24, 25) \rightarrow (12, 2, 4)$	$[11; 2, 2] (11, 24, 26) \rightarrow (15, 1, 5)$	$[11; 2, 3] (11, 24, 27) \rightarrow (12, 4, 2)$
$[11; 2, 4] (11, 24, 28) \rightarrow (13, 2, 3)$	$[11; 2, 5] (11, 24, 29) \rightarrow (13, 4, 3)$	$[11; 2, 6] (11, 24, 30) \rightarrow (13, 3, 2)$
$[11; 2, 8] (11, 24, 32) \rightarrow (15, 4, 2)$	$[11; 3, 1] (11, 25, 26) \rightarrow (12, 3, 4)$	$[11; 3, 2] (11, 25, 27) \rightarrow (12, 4, 3)$
$[11; 3, 3] (11, 25, 28) \rightarrow (16, 1, 5)$	$[11; 3, 4] (11, 25, 29) \rightarrow (15, 5, 1)$	$[11; 3, 6] (11, 25, 31) \rightarrow (14, 2, 3)$
$[11; 3, 7] (11, 25, 32) \rightarrow (15, 2, 4)$	$[11; 3, 9] (11, 14, 23) \rightarrow (9, 3, 2)$	$[11; 4, 1] (11, 26, 27) \rightarrow (12, 3, 2)$
$[11; 4, 2] (11, 26, 28) \rightarrow (13, 2, 4)$	$[11; 4, 4] (11, 26, 30) \rightarrow (17, 1, 5)$	$[11; 4, 5] (11, 26, 31) \rightarrow (14, 3, 4)$
$[11; 4, 6] (11, 26, 32) \rightarrow (14, 4, 2)$	$[11; 4, 8] (11, 15, 23) \rightarrow (10, 2, 3)$	$[11; 4, 9] (11, 15, 24) \rightarrow (10, 5, 1)$
$[11; 5, 2] (11, 27, 29) \rightarrow (13, 4, 2)$	$[11; 5, 3] (11, 27, 30) \rightarrow (16, 5, 1)$	$[11; 5, 4] (11, 27, 31) \rightarrow (14, 3, 2)$
$[11; 5, 5] (11, 27, 32) \rightarrow (18, 1, 5)$	$[11; 5, 7] (11, 16, 23) \rightarrow (10, 4, 3)$	$[11; 5, 8] (11, 16, 24) \rightarrow (11, 3, 2)$
$[11; 5, 9] (11, 16, 25) \rightarrow (11, 4, 2)$	$[11; 6, 1] (11, 28, 29) \rightarrow (13, 2, 3)$	$[11; 6, 2] (11, 28, 30) \rightarrow (14, 3, 4)$
$[11; 6, 3] (11, 28, 31) \rightarrow (14, 2, 4)$	$[11; 6, 4] (11, 28, 32) \rightarrow (15, 4, 3)$	$[11; 6, 6] (11, 28, 34) \rightarrow (19, 1, 5)$
$[11; 6, 7] (11, 17, 24) \rightarrow (10, 3, 2)$	$[11; 6, 8] (11, 17, 25) \rightarrow (11, 5, 1)$	$[11; 6, 9] (11, 17, 26) \rightarrow (11, 4, 2)$
$[11; 7, 1] (11, 29, 30) \rightarrow (14, 4, 3)$	$[11; 7, 2] (11, 29, 31) \rightarrow (17, 5, 1)$	$[11; 7, 3] (11, 29, 32) \rightarrow (15, 2, 3)$
$[11; 7, 5] (11, 18, 23) \rightarrow (10, 4, 2)$	$[11; 7, 6] (11, 18, 24) \rightarrow (11, 4, 2)$	$[11; 7, 7] (11, 29, 36) \rightarrow (20, 1, 5)$
$[11; 7, 9] (11, 18, 27) \rightarrow (13, 3, 2)$	$[11; 8, 1] (11, 30, 31) \rightarrow (14, 4, 2)$	$[11; 8, 2] (11, 30, 32) \rightarrow (15, 3, 2)$
$[11; 8, 4] (11, 19, 23) \rightarrow (11, 2, 4)$	$[11; 8, 5] (11, 19, 24) \rightarrow (11, 2, 3)$	$[11; 8, 7] (11, 19, 26) \rightarrow (12, 5, 1)$
$[11; 8, 8] (11, 30, 38) \rightarrow (21, 1, 5)$	$[11; 8, 9] (11, 19, 28) \rightarrow (12, 4, 3)$	$[11; 9, 1] (11, 31, 32) \rightarrow (17, 5, 1)$
$[11; 9, 3] (11, 20, 23) \rightarrow (11, 3, 4)$	$[11; 9, 5] (11, 20, 25) \rightarrow (11, 3, 2)$	$[11; 9, 6] (11, 20, 26) \rightarrow (12, 4, 3)$
$[11; 9, 7] (11, 20, 27) \rightarrow (12, 2, 3)$	$[11; 9, 8] (11, 20, 28) \rightarrow (12, 4, 2)$	$[11; 9, 9] (11, 31, 40) \rightarrow (22, 1, 5)$
$[11; 10, 3] (11, 21, 24) \rightarrow (11, 4, 3)$	$[11; 10, 4] (11, 21, 25) \rightarrow (11, 4, 2)$	$[11; 10, 5] (11, 21, 26) \rightarrow (12, 2, 4)$
$[11; 10, 6] (11, 21, 27) \rightarrow (13, 5, 1)$	$[11; 10, 7] (11, 21, 28) \rightarrow (13, 4, 2)$	$[11; 10, 8] (11, 21, 29) \rightarrow (12, 3, 2)$
$[11; 10, 9] (11, 21, 30) \rightarrow (13, 2, 3)$		